NAI-WEN CHI

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RESEARCH INTERESTS

Translational biomaterials and drug delivery; tissue repair and regenerative therapies; material-cell interactions; targeted delivery and functional evidence for genetic medicines.

EDUCATION

M.S., Biomedical Engineering / National Tsing Hua University, Taiwan — Feb 2015 - Jan 2017

GPA: 4.05/4.30. Thesis research: nitric-oxide-releasing microparticles for osteoporosis therapy.

- Developed phase-change delivery systems; performed in vitro assays and studies using an ovariectomized rat model. Research contributed to an Advanced Materials publication and patent filings.

B.S., Agronomy / National Chung Hsing University, Taiwan — Sep 2008 - Jun 2012

GPA: 3.25/4.00.

RESEARCH & TRANSLATIONAL EXPERIENCE

R&D and Regulatory Scientist / Coastar Therapeutics, California — Dec 2025 - Present

- Support regulatory gap analysis and the evaluation of development evidence for therapeutic programs.

Research Assistant / Academia Sinica, Taiwan — Apr 2025 - Nov 2025

- Prepared polymer conjugates and ligand-modified nanoparticles; optimized formulation and purification parameters under Dr. Yu-Jung Lin.
- Characterized particle size and zeta potential, quantified drug loading by HPLC, and prepared alginate-based hydrogels with gelation-time evaluation.

R&D Engineer / BioGend Therapeutics, Taiwan — Jan 2018 - Mar 2021

- Contributed to cartilage-repair implant development across scaffold fabrication, preclinical testing, clinical support, and regulatory documentation.
- Supported scaffold fabrication, mechanical testing, mini-pig studies, and surgical usability evaluation.
- Developed tyramine-alginate hydrogel systems for cartilage adhesion and regeneration; co-authored a publication in International Journal of Biological Macromolecules.
- Provided technical support during human clinical surgical cases, including equipment setup and instrument sterilization coordination.

R&D Engineer / Exactech Taiwan Ltd. — May 2017 - Dec 2017

- Supported R&D projects and clinical trials for Class III devices. Exactech Taiwan was acquired by BioGend in January 2018.

PUBLICATIONS

Lin YJ, Chen CC, **Chi NW**, Nguyen T, Lu HY, Nguyen D, Lai PL, Sung HW. In Situ Self-Assembling Micellar Depots that Can Actively Trap and Passively Release NO with Long-Lasting Activity to Reverse Osteoporosis. Advanced Materials. 2018;30(22):1705605. doi:10.1002/adma.201705605.

Wong CC, Lu CX, Cho EC, Lee PW, **Chi NW**, Lin PY, Jheng PR, Chen HL, Mansel BW, Chen YM, Chen CH, Chuang EY. Calcium peroxide aids tyramine-alginate gel to crosslink with tyrosinase for efficient cartilage repair. International Journal of Biological Macromolecules. 2022;208:299-313. doi:10.1016/j.ijbiomac.2022.03.044.

REGULATORY & PROJECT EXPERIENCE

Senior Regulatory Affairs Specialist / Winnoz Technology, Taiwan May 2022 - Jul 2023

- Led TFDA medical-device license applications and amendments; prepared documentation for international IVD submissions.
- Supported ISO 13485 and TFDA quality-management-system audits.

Regulatory Affairs Specialist / Project Manager / BioGend Therapeutics Apr 2021 - May 2022

- Managed TFDA Class I-III medical-device license applications and regulatory documentation.
- Led a TFDA Class II medical-device application and coordinated supporting preclinical evidence.
- Prepared documentation for international medical-device regulatory requirements.
- Supported design controls, ISO 13485 audits, and TFDA QMS audits.

Chief Executive Officer & Project Manager / Wen Fu AI Co., Taiwan Jul 2023 - Mar 2024

- Founded and led a small technology venture, managing online-course and AI-assisted website projects, including scope, schedules, and client communication.

Career Transition & Graduate Study Preparation / Taiwan Mar 2024 - Mar 2025

Prepared doctoral applications and completed English proficiency testing.

TECHNICAL & ANALYTICAL SKILLS

Delivery systems: polymer conjugation and ligand modification; nanoparticle and liposome preparation; scaffold and alginate-hydrogel fabrication.

Cell and animal models: in vitro assays and ovariectomized rat osteoporosis research; support for mini-pig studies.

Biomedical assays: WST-1 viability; alkaline-phosphatase activity; Alizarin Red S mineralization; H&E histology; serum ALP and osteocalcin.

Characterization: DLS and zeta potential; HPLC drug-loading analysis; gelation-time and mechanical testing; sample preparation for NMR and GPC characterization.

Regulatory: regulatory gap analysis and development evidence; TFDA Class I-III; EU MDR/IVDR documentation; ISO 13485; design controls.

SELECTED PATENT FILINGS

Named co-inventor on temperature-responsive drug-delivery and cartilage-repair-device invention families. The statuses below reflect the August 2026 source CV and have not been rechecked for subsequent changes.

- Temperature-Responsive Composition, Method for Fabricating, and Use Thereof: Taiwan invention patent TWI606829B, granted 2017; later lapsed for non-payment of annuities. U.S. published application US20180161276A1, abandoned.
- Cartilage Filling System and Surgical Instrument Set: Taiwan utility model TWM594960U; China utility model CN212730081U.
- Cartilage Volumetric Device and Surgical Instrument Set: Taiwan utility model TWM594437U; China utility model CN212729944U.
- Dry-Type Heating Device: Taiwan utility model TWM590460U; China utility model CN211454393U.

LANGUAGES

Mandarin: native. English: professional working proficiency.